CleanAlert: A Smart GPS-Enabled Real-Time Waste Collection Notification and Community Reward System for Efficient and Responsible Urban Waste Management

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Abstract

In many Indian cities, household waste is collected on fixed routes, but residents rarely know when the collection vehicle will actually reach their street. Missed collections leave waste outside for longer than necessary. This paper presents CleanAlert, an Android application that closes this communication gap. The collector's phone shares its live location through the Global Positioning System (GPS), and when the vehicle reaches a mapped street, residents on that street receive a push notification. After disposing of waste, a resident submits a photograph with location and time, which is checked before points, daily streaks and partner vouchers are awarded. The application uses Google Maps for display and Google's Firebase platform for data synchronization, authentication, photo storage and messaging, with separate Resident, Collector and Admin roles. In prototype testing, the system detected the correct street in 27 of 30 events (90%), delivered notifications in 2.82 seconds on average, passed 22 of 25 disposal audits (88%), and allowed every evaluated user to complete the login and rewards tasks. These results suggest the workflow is technically feasible under the tested conditions, but they come from a small test set without a field deployment, so larger studies are still needed.

Keywords: smart waste management; GPS tracking; geofencing; push notification; Firebase; gamification; mobile application; civic participation

1. Introduction

Efficient solid-waste collection depends not only on vehicles and routes but also on whether residents know when the vehicle will arrive. A published schedule does not show whether the vehicle has reached a particular street. Traffic, route changes and delays open a gap between the planned and the actual collection time. Residents then rely on informal signals, such as hearing the vehicle or asking neighbors, and when the moment is missed, waste stays at doorsteps or roadsides for longer than intended.

CleanAlert targets this last-mile communication gap. It does not replace the municipal collection process; it adds a digital layer on top of it. The collector's location is tracked, residents on the active street are alerted, and a resident can then record disposal with a geo-tagged, timestamped photograph. Points, daily streaks and a locality leaderboard encourage continued participation.

1.1 Related Work

Existing civic-technology research mostly addresses complaint handling. Meshram et al. [1] describe a full-stack civic reporting system that combines maps, authentication and waste-recycling rewards; duplicate detection is left as future work. Prakash et al. [2] combine a ResNet-50/BERT classifier with LSTM-based hotspot prediction for civic issues and report challenges with GPS drift, adoption, multilingual text and misinformation. Naveen et al. [3] propose a multimodal large-language-model framework for issue reporting and validation, noting image quality, hallucination risk, cost, latency and adoption as concerns.

A second line of work studies rewards. Rosenlund et al. [4] used action research and mixed methods to show that gamification can encourage recycling, but with a small sample and possible novelty effects. Satheesh et al. [5] build a TISM/MICMAC and expert-based framework for gamification in solid-waste management, with limited assessment of implementation feasibility. Table 1 summarizes the five studies.

Table 1. Comparison of related works and the gap addressed by CleanAlert.

Ref.	Focus	Approach	Stated limitation / gap	Relation to CleanAlert

[1]	Civic issue reporting, recycle rewards	Full-stack system with maps and authentication	Duplicate detection left as future scope	Similar reward idea; no collection-vehicle arrival alert
[2]	AI civic issue classification	ResNet-50/BERT with LSTM hotspot prediction	GPS drift, adoption, multilingual NLP, misinformation	Shares GPS-drift concern; focuses on reporting, not arrival
[3]	AI-assisted report validation	Multimodal LLM framework (theoretical)	Image quality, hallucination, cost, latency, GPS limits	Validation idea; CleanAlert uses simple location/time checks
[4]	Recycling behavior	Gamified recycling station; action research	Small sample, technical issues, novelty effect	Supports the reward loop; no vehicle-arrival link
[5]	Gamification in waste management	TISM/MICMAC, expert-based framework	Little on feasibility and baselines	Theoretical basis; CleanAlert is an implemented prototype

1.2 Research Gap and Contributions

The reviewed studies concentrate on reporting complaints, classifying issues or rewarding recycling. Little attention is given to linking the real-time arrival of a collection vehicle to a direct resident notification and to verified resident participation. This paper contributes:

- a GPS-based mechanism that tells residents when a collection vehicle reaches their street;
- a notification workflow tied to collector activity through Firebase Cloud Messaging;
- a disposal-submission step that records photograph, location and time;
- a reward loop of points, daily streaks, leaderboards and partner vouchers for verified disposal; and
- an Android prototype, with Resident, Collector and Admin roles, and its initial evaluation on four measures.

2. Methodology

CleanAlert has three user roles. Residents receive alerts and submit disposal evidence; Collectors share their location and trigger street notifications; Admins manage users, routes, submissions and rewards. The Android application follows the Model-View-ViewModel (MVVM) pattern, which separates interface logic from data handling.

2.1 System Architecture

Figure 1 shows the main data paths. Resident and Collector apps exchange data with a Firebase backend that provides authentication with user roles, Cloud Firestore for users, locations, submissions and rewards, Cloud Storage for disposal photos, and Firebase Cloud Messaging (FCM) for notifications. Google Maps supplies map display and address information, and Android device services supply GPS location (FusedLocationProviderClient) and camera capture (CameraX). An Admin panel reads the same backend. Only components used or described in the prototype are shown; municipal-scale analytics remain future work.

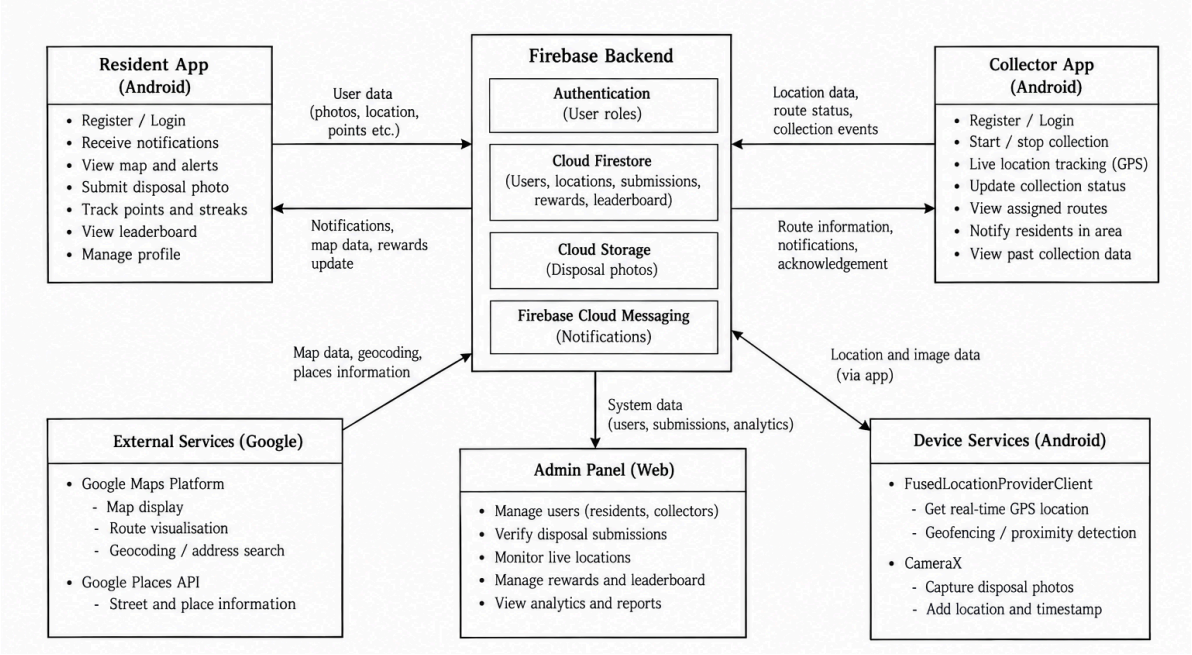


Figure 1. System architecture of CleanAlert.

2.2 Street Detection

The collector's current GPS position is compared with the reference point or boundary of a mapped street. When the distance falls within a configured proximity threshold, that street is treated as the active collection area. This needs no municipal tracking hardware beyond the collector's Android phone. GPS is affected by tall buildings, signal obstruction, device differences and drift, so detection is reported as an observed prototype measurement (correct detections out of test events) rather than a guarantee.

2.3 Operational Workflow

1. The collector logs in, goes online and enables location tracking.
2. The app updates the collector's location along the route assigned by the Admin.
3. The location is compared with mapped street information.
4. When the vehicle reaches a mapped street, the collector triggers the street notification.
5. Residents associated with that street receive a push notification through FCM.
6. The resident disposes of waste and submits a photograph, selecting a waste type (dry, organic, e-waste or other); location and time are attached.
7. The submission is checked against the available location and time information and can be reviewed by the Admin.
8. A verified submission updates the resident's points, daily streak and leaderboard position; points can be exchanged for partner vouchers.

Notification latency is measured from the collector's broadcast to the resident's alert. Verification success is the share of audited submissions that passed. The prototype does not claim to eliminate fraudulent or duplicate submissions, and it separates verification success from behavioral impact: an accepted photograph shows that the workflow ran, not that residents will dispose of waste more responsibly over time.

3. Prototype Implementation

Figures 2 to 4 show the implemented screens. New users register as Resident or Collector; collectors must enter an ID or license number. The Resident home screen shows points and the daily streak, the Submit Waste screen offers camera or gallery photos with waste-type selection and the detected street, and the Rewards screen lists partner vouchers priced in points. The Collector dashboard shows online status, daily totals and the assigned route, and the Collection Map shows the live position on Google Maps. The Admin dashboard offers user management, route assignment, submission review, a rewards catalogue and live Firestore counts. The screenshots were taken on test accounts before any collection or submission data had been entered, so counters read zero.

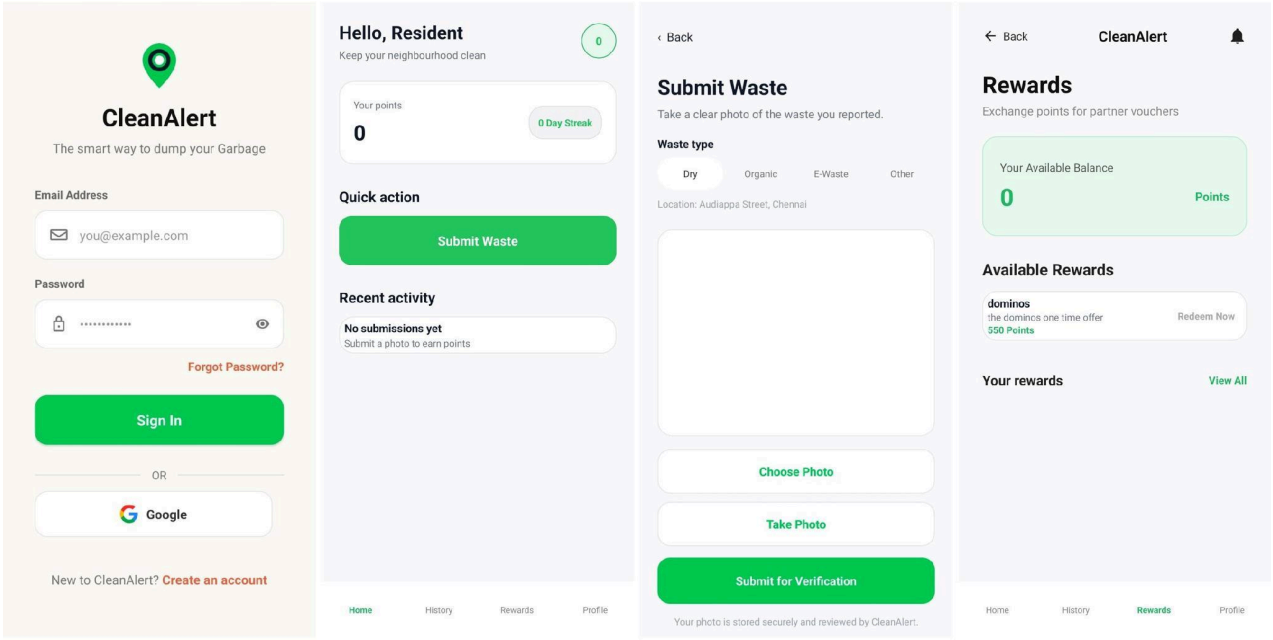


Figure 2. Resident app: sign-in, home with points and streak, Submit Waste, and Rewards.

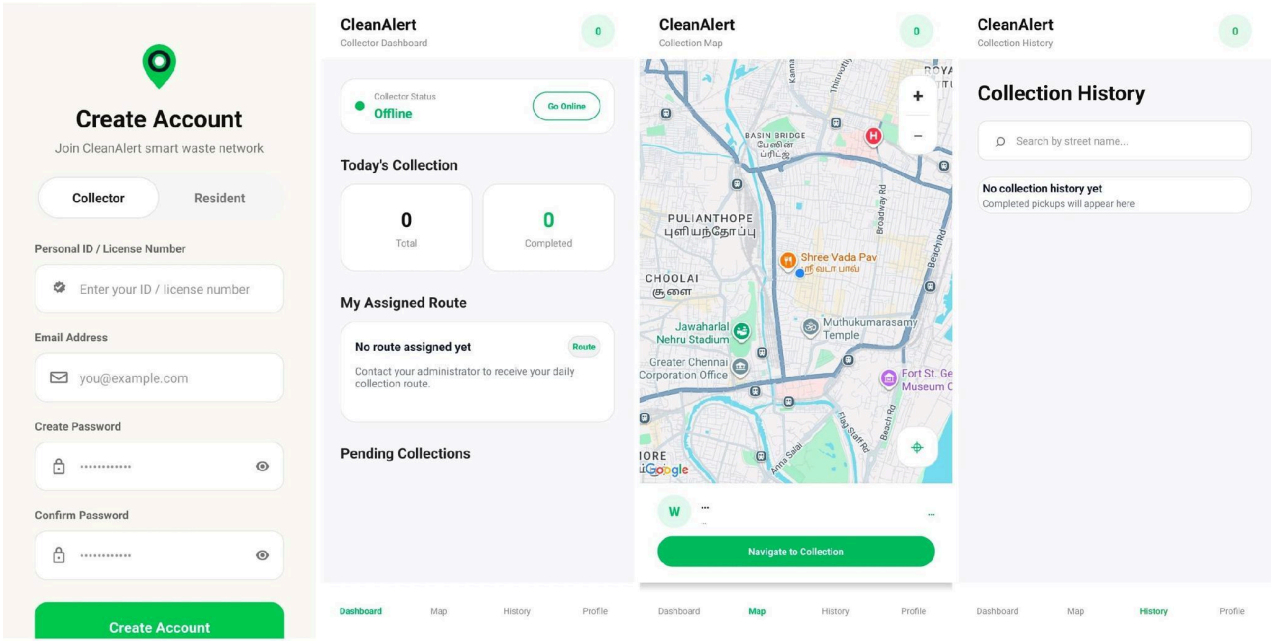


Figure 3. Collector app: account creation, dashboard, collection map and collection history.

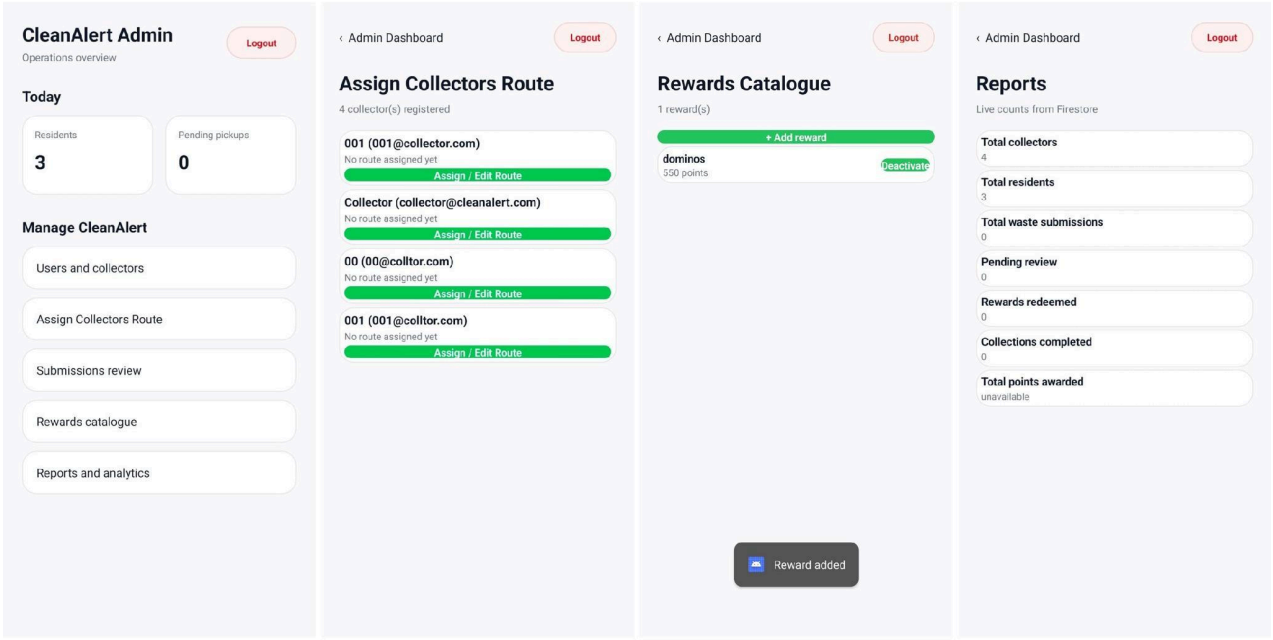


Figure 4. Admin app: operations overview, route assignment, rewards catalogue and reports.

4. Results and Discussion

The prototype was evaluated on four measures: GPS street-detection accuracy, notification latency, disposal-verification success, and completion of selected usability tasks. Table 2 gives the observed values. The 95% intervals were added to show how uncertain a small sample is; they are Wilson score intervals computed from the reported counts.

Table 2. Prototype evaluation results for the CleanAlert workflow.

Metric	Result	Test basis	95% interval	Interpretation
GPS street-detection accuracy	90%	27 correct of 30 events	74–97%	Most tested street detections were correct.
Average notification latency	2.82 s	Collector broadcast to resident alert	not available	Short average delay in testing.
Disposal-verification rate	88%	22 successful of 25 audits	70–96%	Most tested submissions passed verification.
Usability task completion	100%	Login and Rewards tasks	not available	All evaluated users completed the tasks.

4.1 Interpretation

The street-matching workflow worked for 27 of 30 events; three were missed. With only 30 events, the true accuracy could plausibly lie anywhere from about 74% to 97%, so 90% is an initial measurement, not a general accuracy claim. The 2.82-second average latency shows a responsive notification path under the test conditions; real latency will vary with network quality, device state and FCM behavior. Verification passed in 22 of 25 audits (88%); the three failures should be traced in future tests. The 100% task completion applies only to the Login and Rewards tasks, and the number of participants and the wider task set were not specified, so it is not a full usability assessment.

Direct comparison with prior work is limited, because the reviewed systems [1]–[5] do not report arrival-notification latency or street-level detection accuracy. What can be said is that CleanAlert addresses a different event (vehicle arrival) and, like [2] and [3], is exposed to GPS drift. The failed detections and audits point to concrete follow-up checks: GPS drift, street boundaries, image metadata, network conditions and the verification logic.

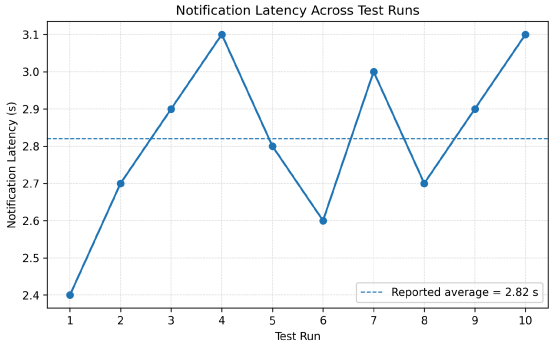


Figure 3. Notification latency across illustrative test runs.

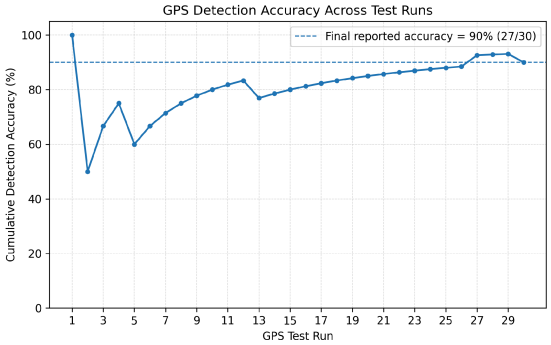


Figure 4. GPS detection accuracy across illustrative test runs.

4.2 Limitations

- The test set is small and no municipal field deployment was carried out.
- Only aggregate results were kept; per-run logs were not available, so no per-run plots or variance are reported, and a false-notification rate was not measured.
- GPS accuracy varies with signal obstruction, urban density and device characteristics; notification delivery depends on connectivity and notification settings.
- The 88% verification rate reflects the audited samples and is not a general measure of verification accuracy; duplicate and fraudulent submissions are not handled.
- The effect of points, streaks and leaderboards on behavior has not been measured, and the Admin module lacks municipal-scale analytics and multilingual support.

5. Conclusion and Future Work

CleanAlert links live collector GPS tracking, street-level push notifications, photo-based disposal evidence and a reward loop in one Android workflow. Prototype tests showed 90% street-detection accuracy over 30 events, 2.82 s average notification latency, an 88% verification rate over 25 audits, and full completion of the evaluated login and rewards tasks. These are preliminary signs of technical feasibility, not an impact evaluation.

Future work includes: a field pilot in one municipal ward under real GPS drift; measuring end-to-end latency and false-notification rate; diagnosing the failed detections and audits; a controlled study of rewards against no rewards; duplicate-submission detection; route analytics for Admins; multilingual support; and long-term retention studies.

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